★ Assignment

Complete the following program. Submit your **.py** file by the next class.

Student Information System

```
Build a menu-driven Python program using a dictionary.
Key = Student ID (int), Value = nested dict with 'name' and 'marks'.
```

#	Feature	Description
1	**Add Student**	Prompt for student ID, name, and marks. Store in the dictionary. Handle duplicate IDs gracefully.
2	**View All Students**	Loop through the dictionary and display every student's ID, name, and marks neatly.
3	**Update Marks**	Ask for a student ID, check if it exists, then allow the user to enter new marks.
4	**Search Student**	Ask for a student ID and display that student's full details, or show 'Student not found'.
5	**Find Topper**	Iterate all students, compare marks, and display the student with the highest marks.

Starter Structure (fill in the logic):

```python
students = {}   # student_id: {'name': ..., 'marks': ...}

while True:
    print("\n--- Student Information System ---")
    print("1. Add Student")
    print("2. View All Students")
    print("3. Update Marks")
    print("4. Search Student")
    print("5. Find Topper")
    print("6. Exit")
    choice = input("Enter choice: ")

    if choice == "1":
        # TODO: get id, name, marks and add to students dict
        pass
    elif choice == "2":
        # TODO: loop and print all students
        pass
    # ... continue for choices 3-5
    elif choice == "6":
        break
```

■ Bonus Challenges (optional)

Bonus 1: Validate that marks are between 0 and 100.

Bonus 2: Sort and display students by marks (highest first).

Bonus 3: Calculate and display the class average marks.

Good luck! Remember: a dictionary is your best friend when data has a name.